

MAYANK GANGREDIWAR

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Pune, Maharashtra - 411044, India

OBJECTIVE

AI/ML developer with expertise in microservices automation, multimodal transformer architectures, and production ready application development. I build scalable AI systems, optimize inference pipelines, and deliver end-to-end engineering impact across cloud, mobile, and web platforms. I bring a strong product mindset evaluating features from both technical and business perspectives to drive user value, product growth, and long-term organizational success.

EDUCATION

Bachelor of Technology, Electronics and Telecommunications, PCCOE, Pune	(Expected 2026, CGPA 7.67)
Honors in Deep Learning, PCCOE, Pune	(2023 - 2026, CGPA 9.0)
Indian School Certificate, Maharshi Jiwatode Junior College, Chandrapur	(2020 - 2022, 83.83%)

SKILLS

Programming Languages: Python, SQL, C++, C, MATLAB, Java, Dart

Machine Learning & AI: Deep Learning, NLP, Model Fine-Tuning, LLM Integration, Prompt Engineering
Frameworks: PyTorch, TensorFlow/Keras, Streamlit, Jupyter, Flutter

Cloud & DevOps: Google Cloud, Firebase, GCP Functions, GitHub Actions (basic)

Tools & APIs: OpenAI APIs, Claude API, Gemini API, Deepgram, Eleven Labs, pyBigWig, pybedtools, OpenPyXL, GitHub

Soft Skills: Product Thinking, Rapid Learning, Leadership, Collaboration, Ownership, Analytical Reasoning, Communication

WORK EXPERIENCE

AI Intern - Microservices, Product Development & Applied Machine Learning Asymmetrical Learning	Jun 2025 – Nov 2025 304 KBC Tower, Karve Nagar, India
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- Engineered and automated core AI microservices including Audio Generation Engine and AI-Powered PDF Reader using OpenAI, Gemini, Claude, Eleven Labs, and Deepgram APIs.
- Optimized NLP, OCR, and TTS pipelines, improving inference stability, reducing latency, and contributing to model fine-tuning.
- Co-developed the Wriddles App: built Flutter frontend, integrated Firebase backend, and implemented serverless logic with Google Cloud Functions (GCP).
- Designed and deployed the official Wriddles website (wriddles.com) and supported Play Store & App Store release processes.
- Built scalable automation pipelines, improved cloud deployments, and collaborated directly with the CTO on product and system design decisions.

PROJECTS

Wriddles Mobile Application (for Asymmetrical Learning)

Tech: Flutter, Firebase, Google Cloud Functions (GCP), Python, OpenAI APIs



- Co-developed a gamified riddles learning app using Flutter & Firebase, contributing to Play Store & App Store deployment.
- Implemented an AI-driven personalization engine using explore-exploit algorithms for adaptive difficulty.
- Built automated pipelines for riddle tagging and metadata enrichment, improving scalability and reducing manual workload.
- Developed and deployed the official website (wriddles.com), enabling public platform access and seamless navigation.

AI-Based Smart Water Quality Analyzer & Disease Outbreak Forecasting System

Tech: ESP32, IoT Sensors, Python, Random Forest, LSTM, HTML, CSS, JS



- Developed an IoT-enabled water quality monitoring system using ESP32 with pH, TDS, turbidity, DO, & temperature sensors.

- Built ML models for WQI prediction ($R^2 = 0.95$), contamination detection (97.7%), and degradation forecasting using LSTM.
- Created a dataset of 48k+ synthetic & real readings with noise augmentation & temporal slicing for robust training.
- Developed a real-time dashboard (HTML, CSS & JS) for live sensor data visualization, predictions, alerts, and trend analytics.

Multi-Modal Transformer for Chromatin Accessibility Prediction



Tech: PyTorch, Transformers, ENCODE, pyBigWig, pyfaidx, Genomics

- Built a multi-modal transformer integrating DNA sequence embeddings with functional genomic signals (histone marks, TF binding).
- Preprocessed ENCODE datasets using pyBigWig, pybedtools, and pyfaidx to produce fused multimodal inputs.
- Implemented cross-modal attention layers and trained the model with masked-accessibility objectives for regulatory prediction.
- Achieved higher ROC-AUC/PR-AUC vs. sequence-only baselines and extracted biologically meaningful motifs via attention maps.

ADDITIONAL INFORMATION

Publications (Accepted):

- **“AI-Based Smart Water Quality Analyzer and Disease Outbreak Forecasting System”**
 Mayank Gangreediwar, Riddhi Kulkarni, Abhijit Kshirsagar
 IEEE PuneCon 2025 - Accepted for Presentation & Publication
 To be presented in December 2025

Competitions:

- Smart India Hackathon (SIH 2025) - Shortlisted for Round 2
- AI for Climate Change Hackathon (2025) - Shortlisted for Round 2
- Avishkar Research Competition - Sustainable Innovation Track
- River ReThink (2025) - Shortlisted for Round 2
- SPECTRUM (State-Level Tech Event) - Finalist

Leadership & Achievements:

- Overall Advisor, ETSA PCCOE (2025–2026)
- Vice President, ETSA PCCOE (2024–2025)
- Co-Convener, Baby Conference - First-Year Department, PCCOE
- Overall Coordinator, Research Paper Conference - Abhiyantrix'25
- Former Vice-Captain, Chandrapur District Cricket Team
- Member, PCCOE Basketball Team - Inter-college Representation

Personal Interests:

- Basketball (3× Inter-department Winner)
- Cricket (Professional - 3+ years)
- 3×3 Speedcubing (Best: 16.59s)
- Reading
- Stock Market Analysis & Finance